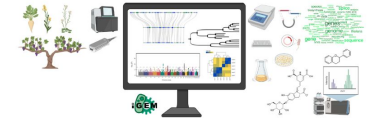


Prof. Dr. Boas Pucker

Long Read Genomics

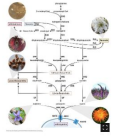


- Biochemistry at HHU Düsseldorf
- (Systems) Biology at Bielefeld University
- Doctoral student (CeBiTec, Bielefeld University)
 - Genomics & Bioinformatics; synthetic biology (iGEM)
- Post doc (Ruhr-University Bochum)
- Post doc (Department of Plant Sciences, Cambridge, UK)
- Plant Biotechnology & Bioinformatics, TU Braunschweig (2021-2025)
 - Specialized plant metabolites, applied bioinformatics
- University of Bonn (since 2025): Plant Biotechnology & Bioinformatics



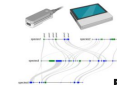
Specialized Metabolism

Plants produce a plethora of specialized metabolites. These allow plants not only to cope with numerous environmental challenges, but they can also have beneficial effects on humans. Many plant species have been successfully used in traditional medicine. Investigations of specialized plant metabolism can reveal the biosynthesis pathways and enable heterologous production of drug candidates. Our fundamental research on biosynthetic pathways paves the way for translation into industrial applications.



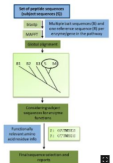
Plant Genomics

Plant genomes harbour genes, which are the blue prints for enzymes involved in various biosynthetic pathways. Knowledge about plant genomes unravels the mysteries of the specialized biosynthetic pathways in plants. Rapid developments in long read sequencing technologies enable us to study even large and complex plant genomes.

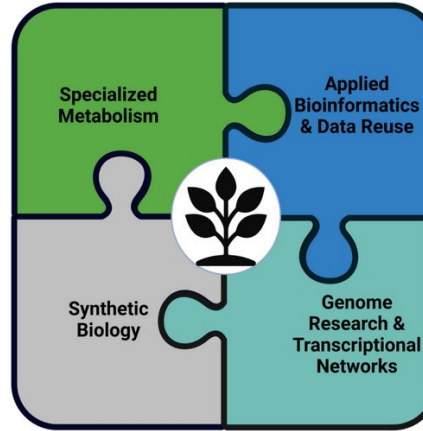


Bioinformatics

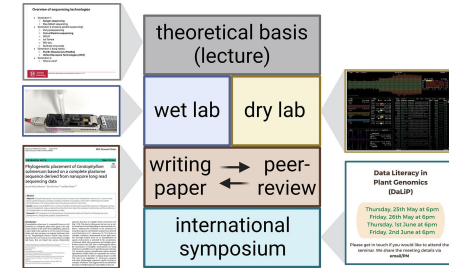
Specific biological questions can be answered through sophisticated tools. Automatic analysis of large data sets and the integration of genomic, transcriptomic, and metabolomic data are often required. Applications include the discovery of biosynthetic pathways and their regulators, the detection of biosynthetic gene clusters, and the identification of tolerance/resistance mechanisms. Bioinformatic tools are an effective way to generate hypotheses and to guide molecular biology experiments.



- Flavonoids: Anthocyanins, Flavonols, Flavones, Proanthocyanidins
- Withanolides
- Carotenoids
- ...



Tools: KIPES, NAVIP, MGSE, MYB/bHLH_annotator, DupyliCate...



Availability of slides

- All materials are freely available (CC BY) - after the lectures:
 - GitHub: <https://github.com/bpucker/LRG>
- Questions: Feel free to ask at any time
- Feedback, comments, or questions: [pucker\[a\]uni-bonn.de](mailto:pucker[a]uni-bonn.de)



My figures and content can be re-used in accordance with CC BY 4.0, but this might not apply to all images/logos.

Background

Discovery of DNA

- Isolation of DNA by Friedrich Miescher in 1869
- Working in the castle in Tübingen
- Used a kitchen as lab



Friedrich Miescher
(1844-1895)

Discovery of DNA structure

- WATSON, J., CRICK, F. Molecular Structure of Nucleic Acids: A Structure for Deoxyribose Nucleic Acid. *Nature* **171**, 737–738 (1953).
<https://doi.org/10.1038/171737a0>

MOLECULAR STRUCTURE OF NUCLEIC ACIDS

A Structure for Deoxyribose Nucleic Acid

WE wish to suggest a structure for the salt of deoxyribose nucleic acid (D.N.A.). This structure has novel features which are of considerable biological interest.



What is the genome of a plant?

Major types of DNA in plants

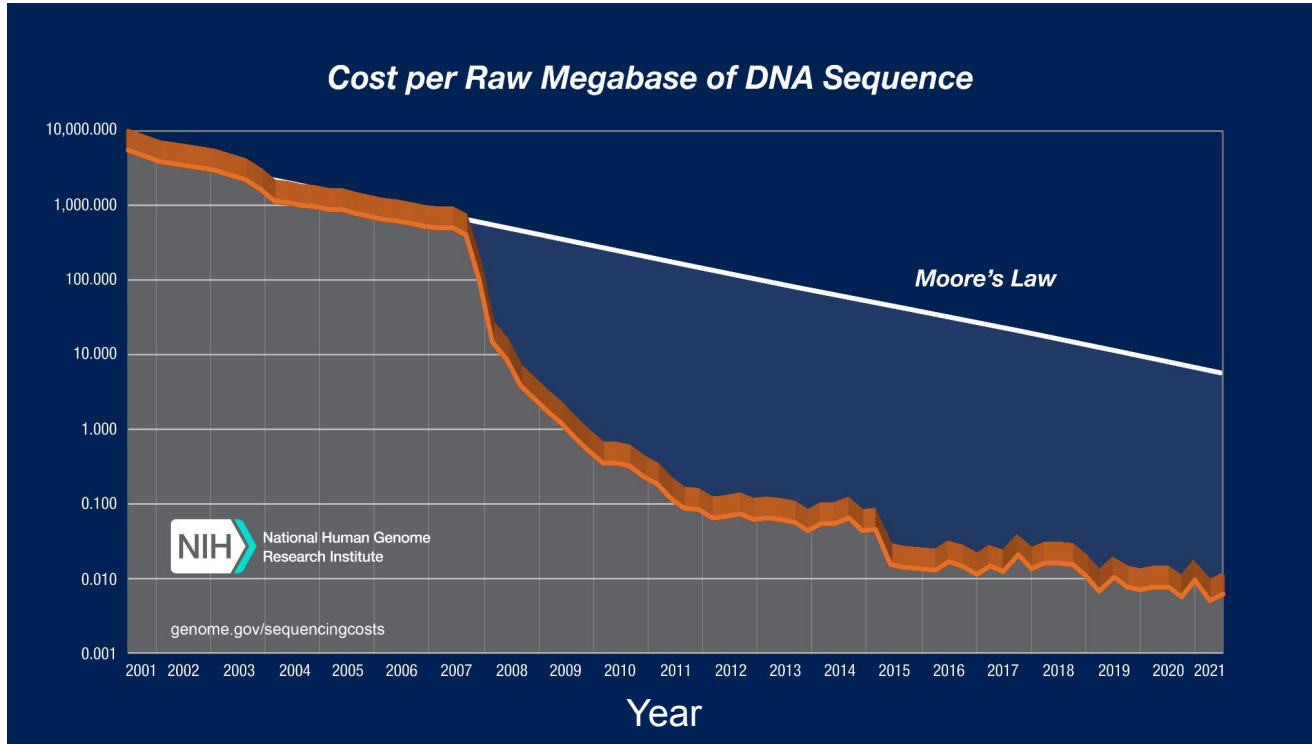
- gDNA from the nucleus
- mtDNA from the mitochondria (chondrome)
- cpDNA from the chloroplast (plastome)
- pDNA (plasmids, only in biotechnological applications)

Why do we sequence plant genomes?

- Primer design (banana)
- Assessment of genetic diversity (population genomics)
- Trait discovery (yam)
- Breeding (sugar beet)
- Understanding the evolution (At7)



Development of sequencing capacity



Do you know sequencing methods?

Sequencing technologies (selection)

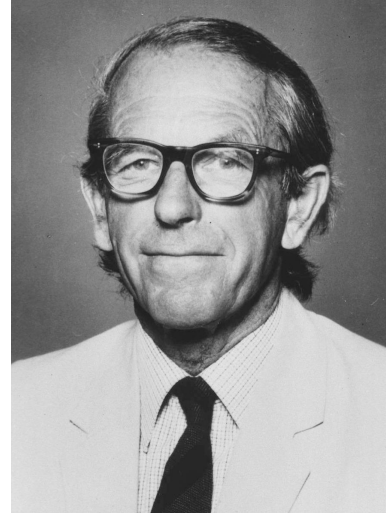
- Generation 1:
 - **Sanger sequencing**
 - Max-Gilbert sequencing
- Generation 2 (massive parallel sequencing):
 - 454 pyrosequencing
 - **Solexa/Illumina sequencing**
 - SOLID
 - Ion Torrent
 - BGI-seq
 - Synthetic long reads
- Generation 3 (long reads):
 - **Pacific Biosciences (PacBio)**
 - **Oxford Nanopore Technologies (ONT)**
- Generation 4:
 - What is next?

Sanger sequencing

Inventor of Sanger sequencing

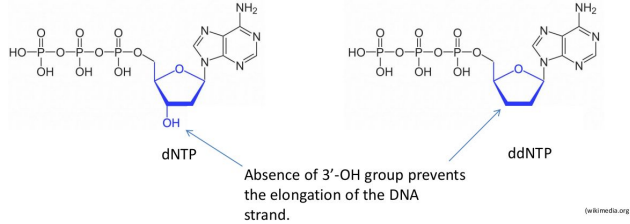
Nobel prizes for:

- 1) Protein sequencing (1958)
- 2) DNA sequencing (1980)



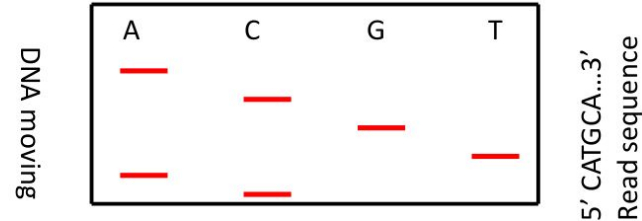
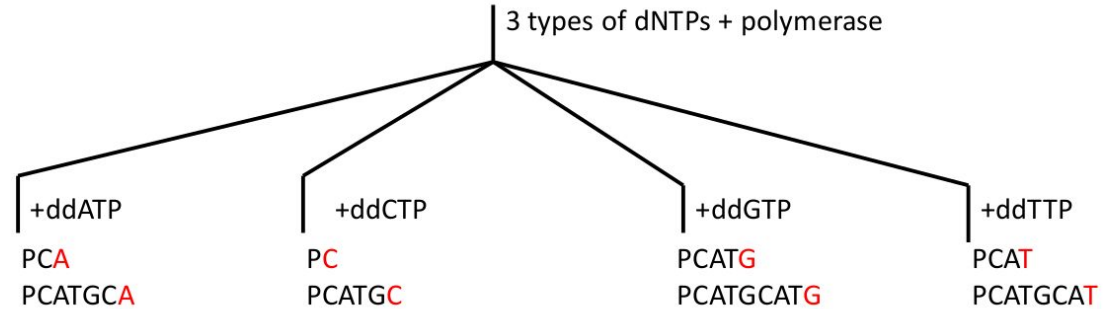
Frederick Sanger
(1918-2013)

Concept of Sanger sequencing



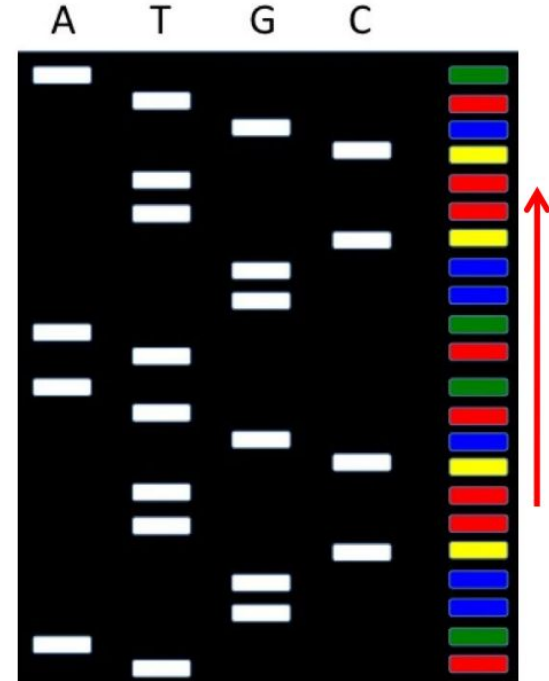
Primer (P):
Template:

5' –TGCATGGCATGATGCATG–3'
3' –ACGTACCGTACTACGTACGTACGTACGTCTAGGT–5'



Sanger sequencing - original version

Two persons analyze the gel: one is calling the base ('basecaller') and the other person is writing down the bases



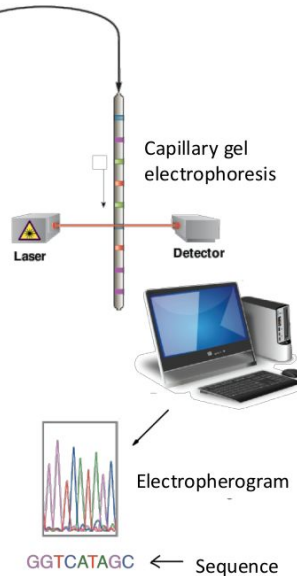
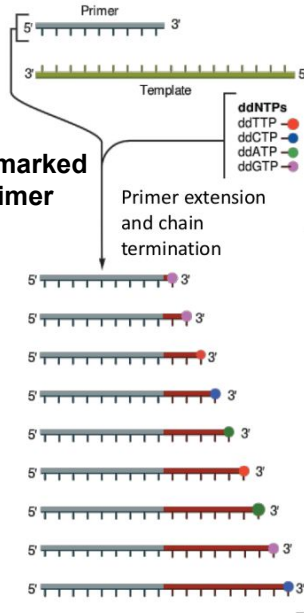
(modified from wikipedia.org)

Sanger sequencing - today

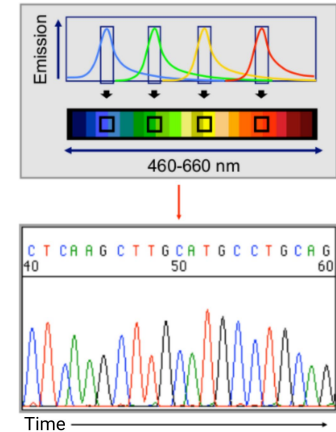
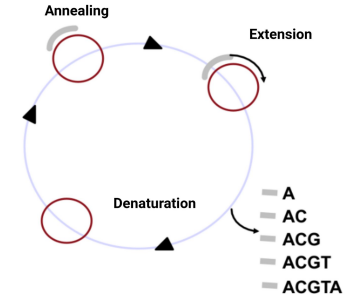
Only one reaction!

ddNTPs are marked instead of primer

Low input required due to cycle sequencing

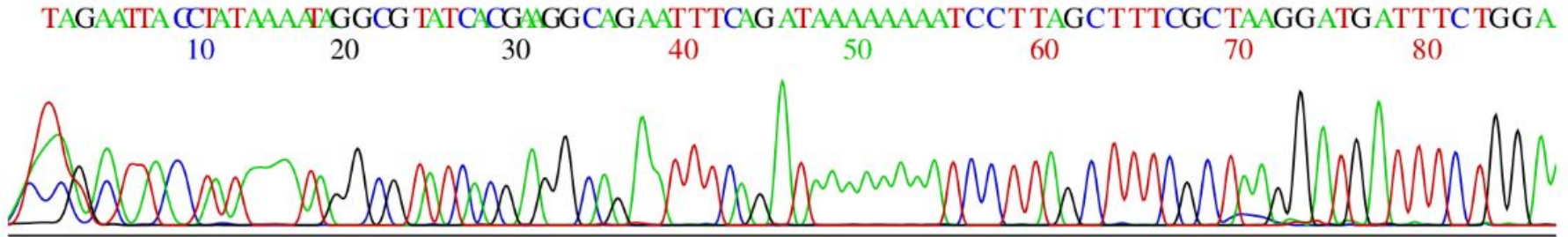


Capillary avoids interference of adjacent lanes on a gel!



TRACE (.abi/.ab1)

- Original result file of ABI basecaller (Sanger sequencing)
- Contains only one read per file



- There are two types of lines: header and sequence
- Header line starts with '>'; can contain name and information about sequence
- Example:

```
>seq1 len=5
```

```
ACGTA
```

```
>seq2 len=10
```

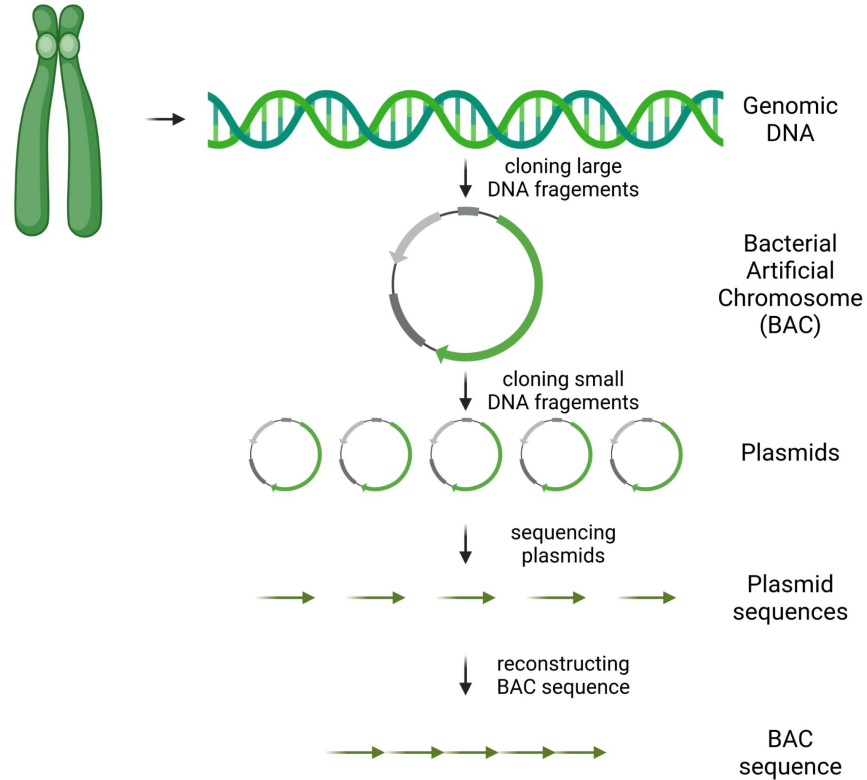
```
ACGTA
```

```
ACGTA
```

```
>seq len=1
```

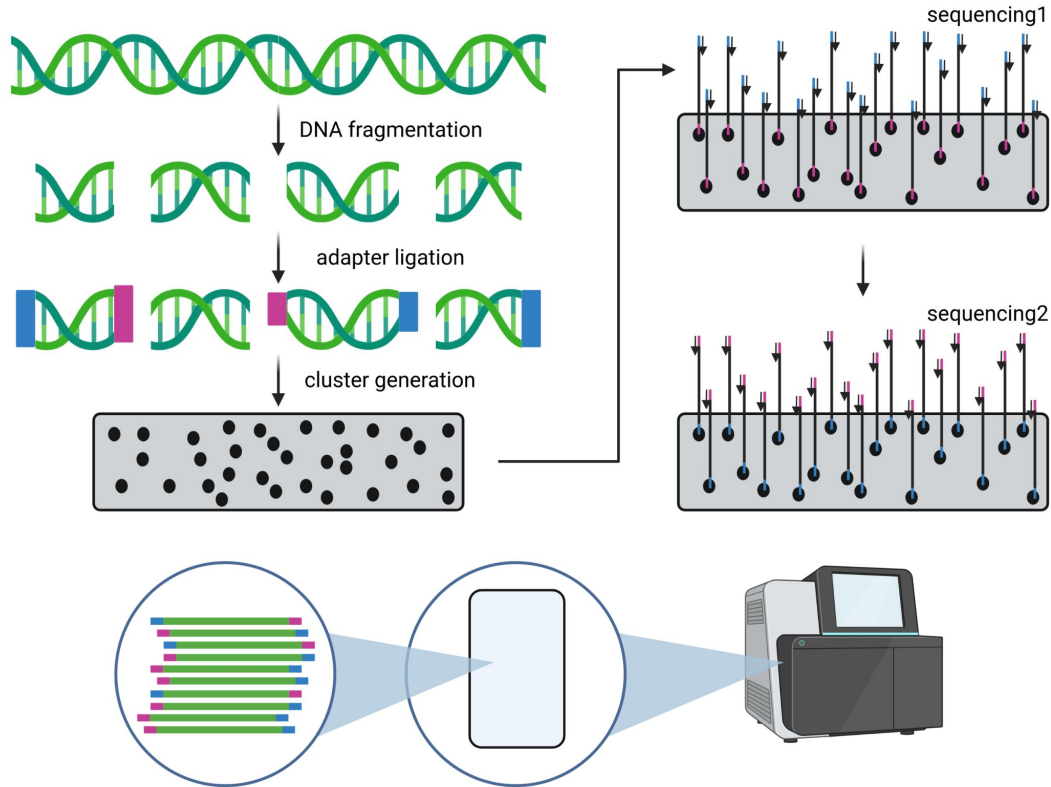
```
A
```

BAC-based sequencing strategy



Illumina sequencing

Illumina sequencing



Phred-Score

- Negative logarithm of the error probability for given position in read
- Multiplication by 10 to avoid floats
- Encoding quality value as single character (ASCII)

Phred quality score	Error probability	Accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1,000	99.9%
40	1 in 10,000	99.99%
50	1 in 100,000	99.999%

- Standard format for sequences with associated quality information
- Four lines per entry:
 - Header starts with @ (title + description)
 - Sequence
 - + (optional repetition of header)
 - Quality (phred encoded in ASCII character)
- Different versions exist that use different quality values offsets
- Example:

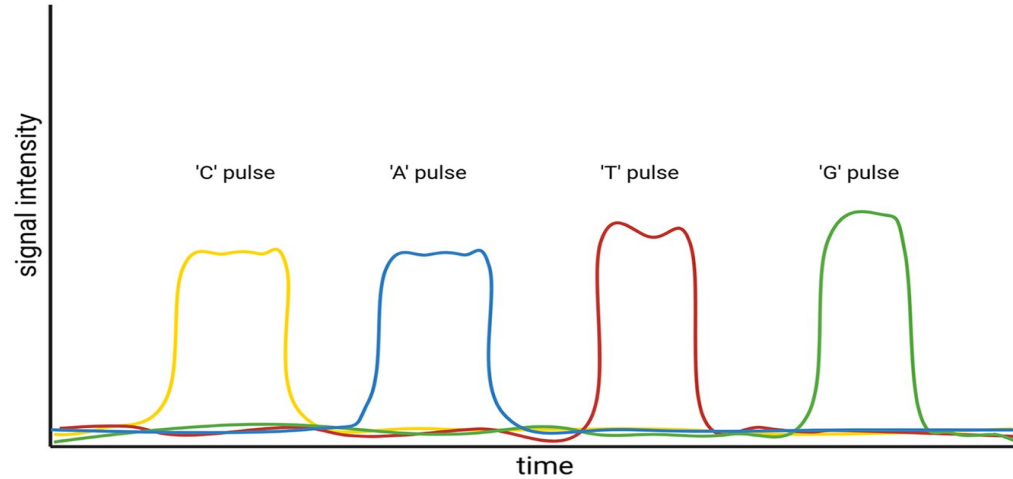
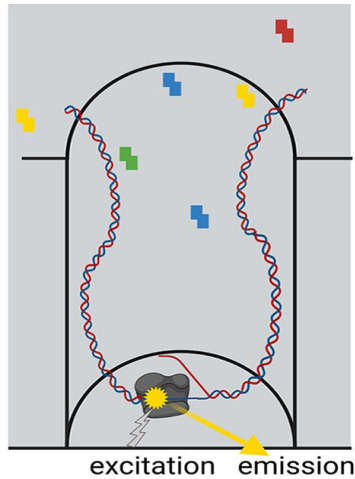
@seq1

ACGTACGTACGT

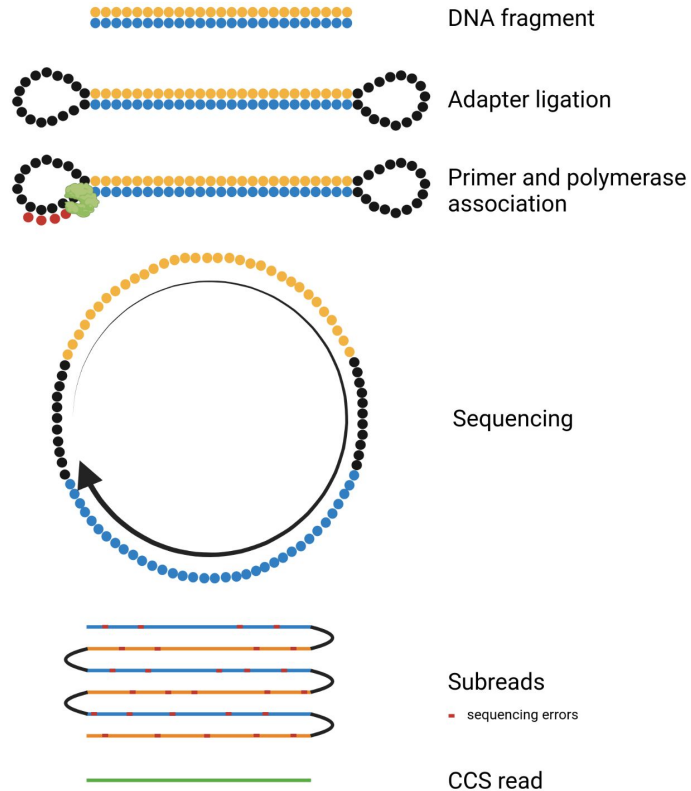
+

""?CB"" :DC"

- Polymerase located at bottom of well
- ZMWs = Zero Mode Waveguide

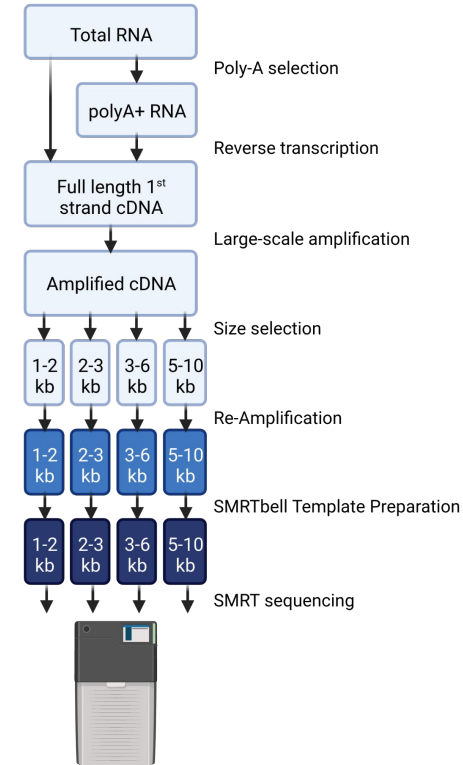


PacBio - HiFi



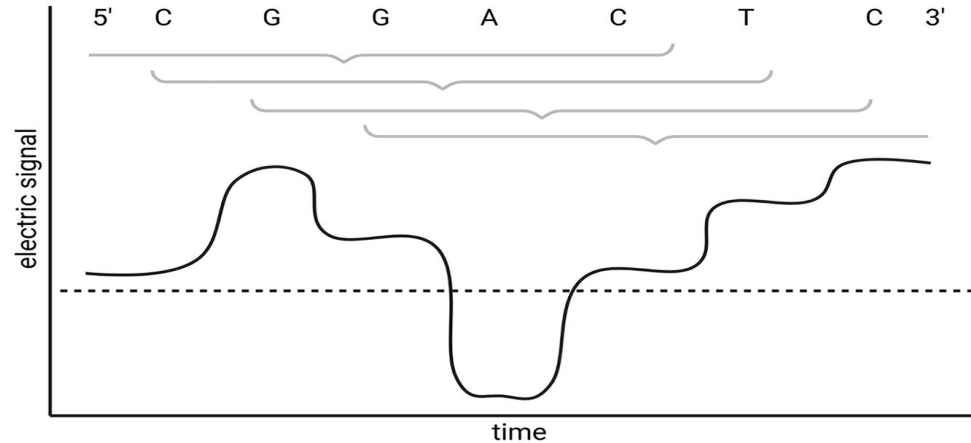
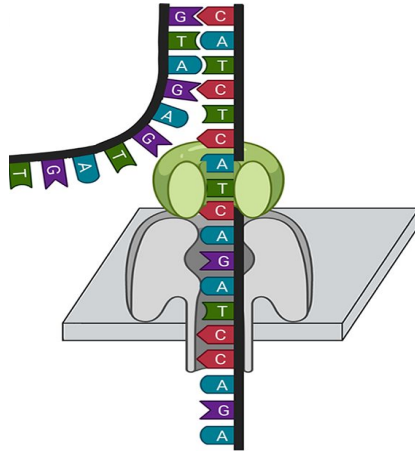
PacBio Iso-Seq (cDNA sequencing)

- SMRT = Single Molecule Real Time sequencing
- Full length cDNA sequencing is beneficial for gene prediction
- Iso-Seq generates several kb long reads and not only 2x300bp



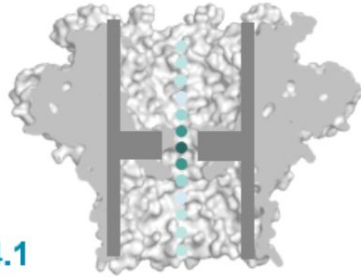
Under development since the 1980s

“Analysis of anything, anywhere by anyone” ... not restricted to DNA sequencing

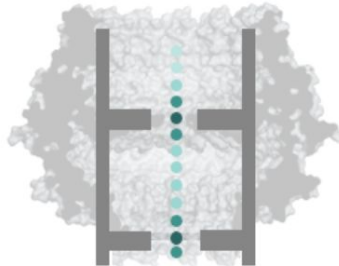


Nanopores

R9.4.1



R10

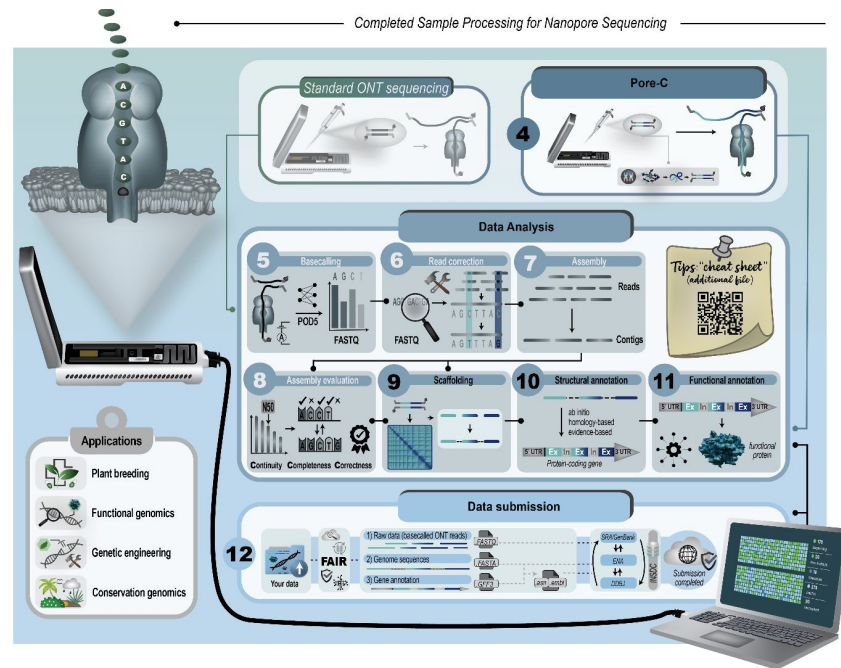
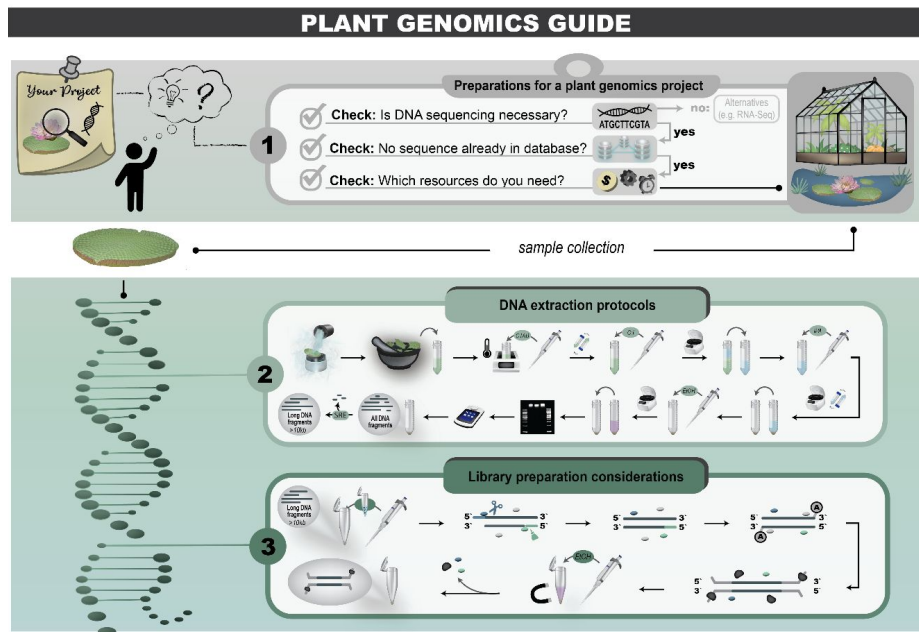


ATCGGAAAAAAAAAATCACGCCACGTCCAAA



Plant genome sequencing workflow

Plant Genomics Workflow



Supplementary file with all commands!

ONT sequencing workflow

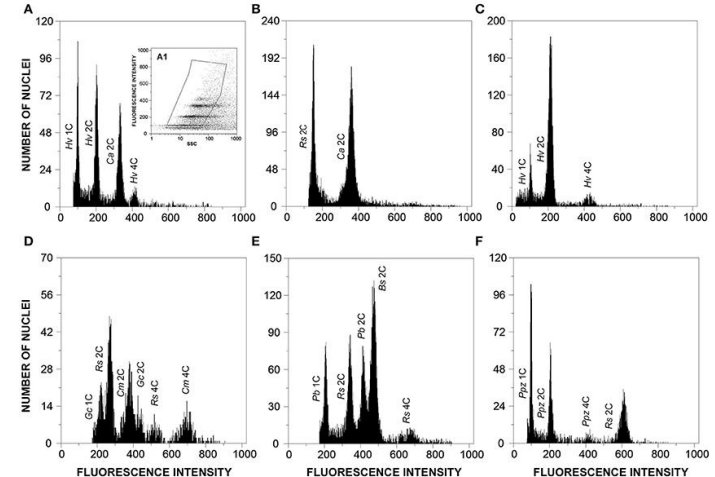
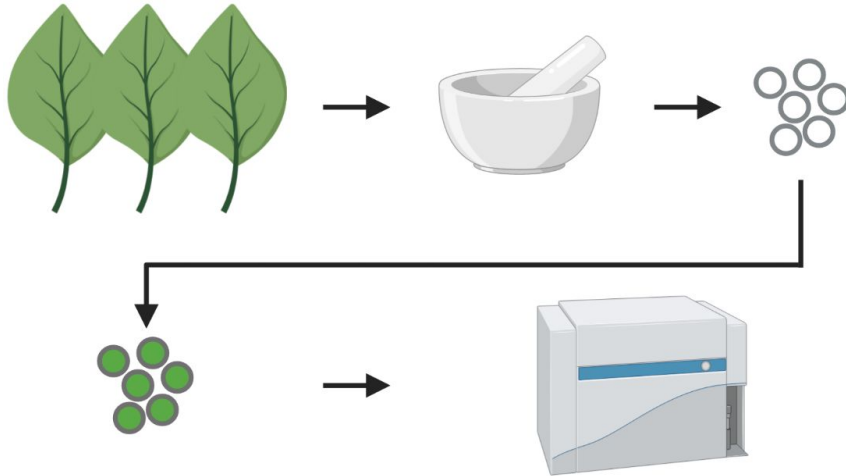
		task	consumed time	hands-on time	equipment	estimated costs of consumables	estimated costs of lab equipment
A		plant incubation in darkness	2-3d	1h			
B		↓ non-destructive sampling	-	1h			
C		↓ DNA extraction	1d	8h	waterbath, centrifuge	\$50	\$1000
D		↓ quality control	1h	1h	NanoDrop, Qubit	\$20	\$8000
E		↓ short fragment depletion	2h	1h	centrifuge	\$50	
F		↓ quality control	1h	1h	NanoDrop, Qubit	\$20	\$5000
G		↓ library preparation & sequencing	1-5d	4-16h	centrifuge, magnetic rack, sequencer	\$3000	\$5000
H		↓ basecalling	1d	1h	computer with GPU		\$250
I		↓ assembly	1-15d	1h			\$1000
J		↓ polishing	1-5d	1h	compute cluster / cloud		\$3000
K		↓ annotation	1-5d	1h			
L		↓ data submission	2h	2h	fast internet connection		

Considerations & preparations

- What is the expected/estimated genome size?
- What is the ploidy of the species?
- Which individual should be sequenced?
- Which plant parts are suitable for DNA extraction?
- What materials are needed for DNA extraction and sequencing?

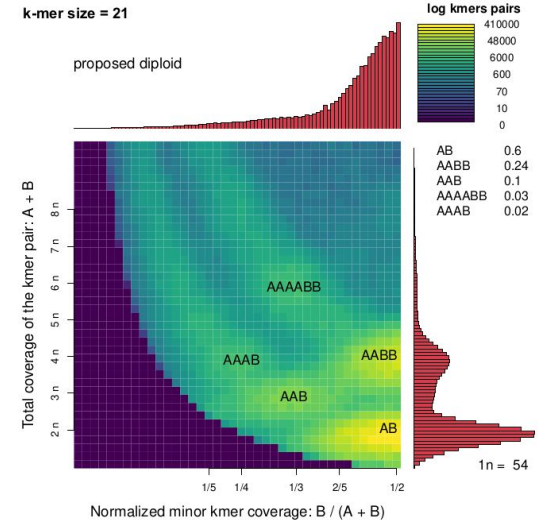
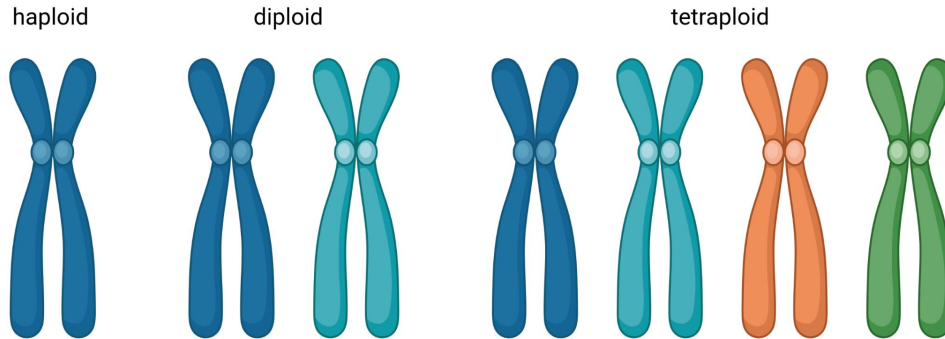
Genome size

- Genome sequences of closely related species
- Flow cytometry is used to measure genome size biochemically
 - C-value
- Databases for plant genome sizes: <https://cvalues.science.kew.org/>



Ploidy

- Ploidy = copy number of same chromosome
- Many plants are polyploid
- Some polyploid plants have close diploid relatives
- Smudgeplots to analyze polyploidy based on short reads

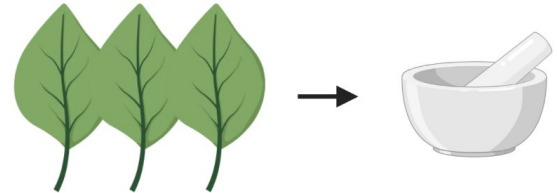


Picking an individual

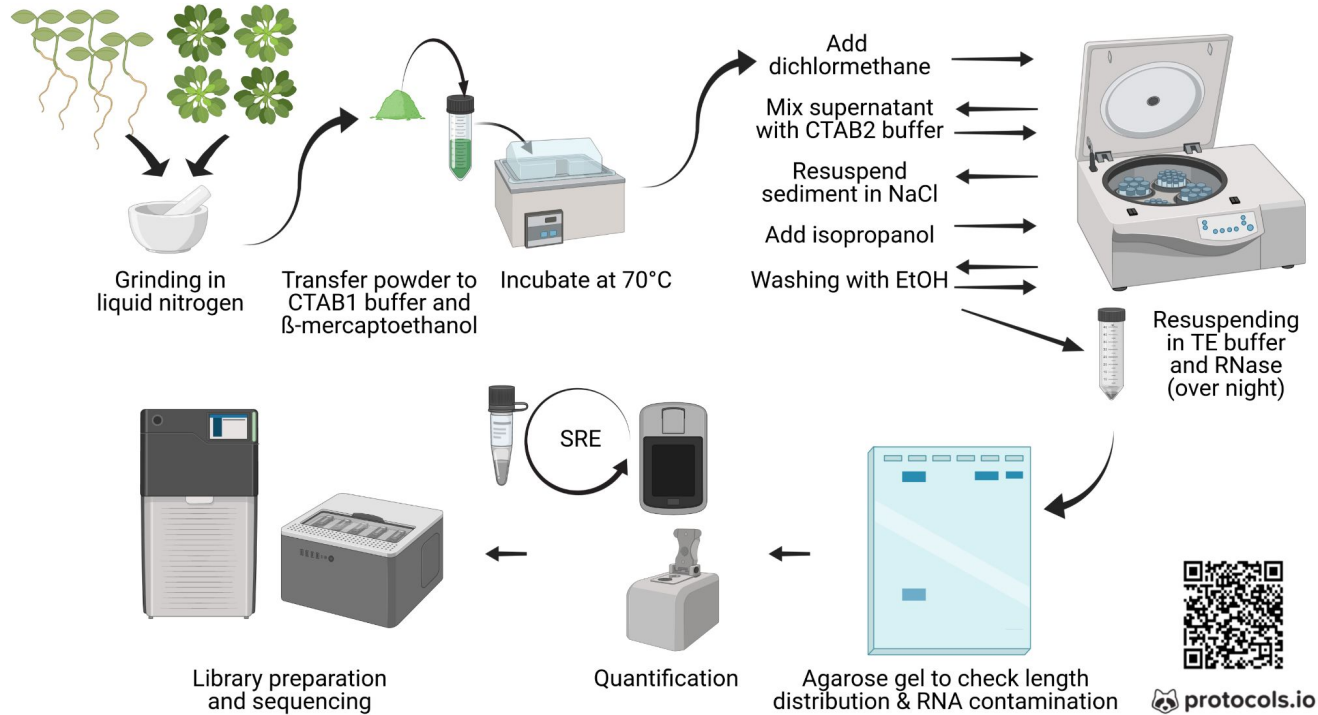
- Plant should survive the sampling for DNA extraction
- Plant should be a good representation of the species
- Source of the plant is important:
 - Restrictions through material transfer agreements (MTAs)?
 - Restrictions through Nagoya protocol / Access and Benefit Sharing (ABS) law

Plant part for DNA extraction

- Young leaf is often a good choice
- Small cells result in higher density of nuclei per weight
- Concentration of specialized metabolites should be low
- Amount of sugar should be low
- Amount of chloroplast should be low
- Sample should not be contaminated with bacteria/fungi

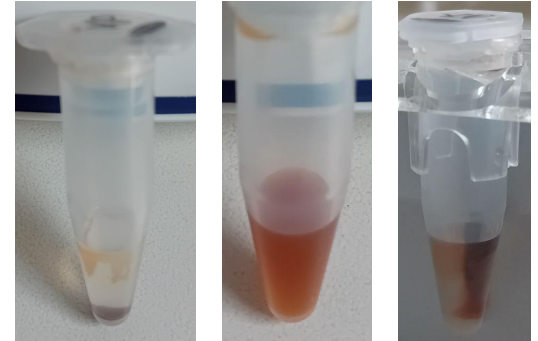


DNA extraction workflow



Material required for sequencing

- DNA repair (companion module)
- Magnetic beads for purification
- Library preparation kit
- Sequencing kit
- Wash kit with DNase



ONT sequencing devices

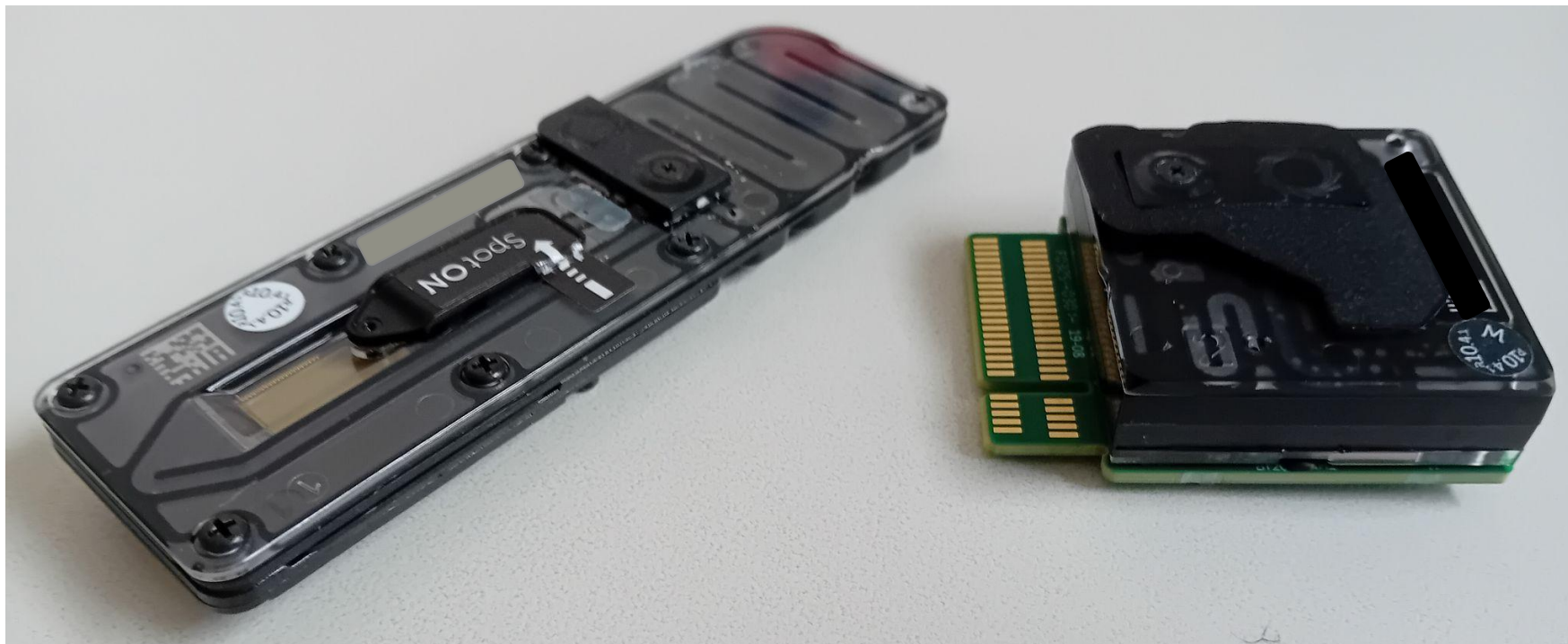


MinION



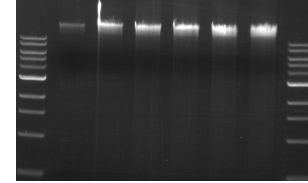
PromethION 2 Solo

Flow cells: MinION vs. PromethION



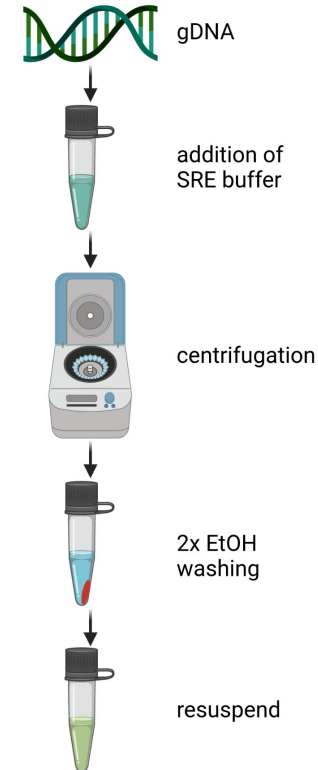
Quality control

- Agarose gel electrophoresis
- Photometric measurement via NanoDrop
- Quantification with Qubit



Short Read Eliminator (SRE)

- Proprietary salt mix for DNA precipitation
- Removal of <10kb DNA fragments
- Depletion of <20kb DNA fragments
- ONT read length distribution can be substantially improved



DNA repair

- Repairing single strand DNA breaks
- Repairing DNA ends (3'-A overhang required for adapter ligation)

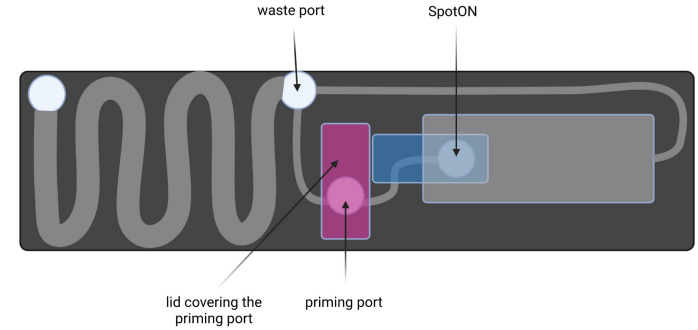


Library preparations

- Repaired DNA is subjected to library preparation
- Addition of adapters to DNA fragments (expensive ligase)
- Concentration of DNA can be quantified via Qubit measurement (optional)
 - Control step to ensure that library construction is working
- Purification of DNA with magnetic beads

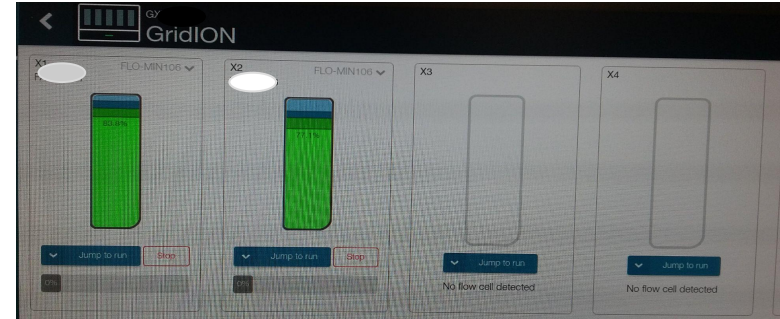
Loading flow cell

- Removal of storage buffer
- Priming of flow cell
- Introduction of air bubbles must be avoided!!!
- Fully open ports are crucial to inject solutions (avoid force)
- Video tutorial: <https://www.youtube.com/watch?v=Pt-iaemrM88>



Monitor sequencing

- Number of active nanopores can be monitored in real time
- Output is estimated in real time
- Read length distribution can be assessed
- Speed of the sequencing can be monitored
- Quality of the reads is displayed



Nanopore activity

- Status of nanopores on flow cell
- Light green is good (sequencing)
- Number of sequencing nanopores change over time
- Air bubbles can destroy nanopores



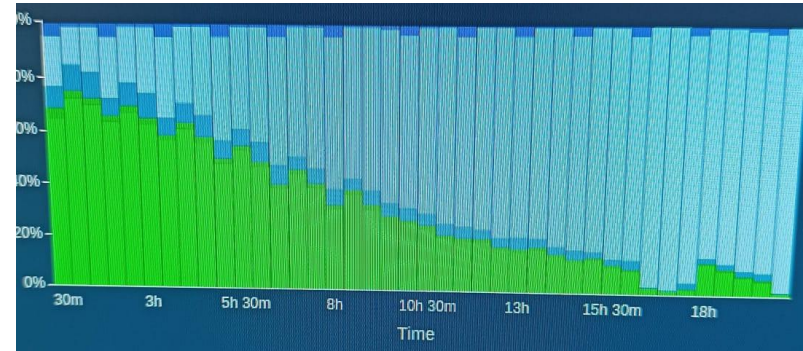
Read length distribution

- Read length distribution
- N50 summarizes length distribution



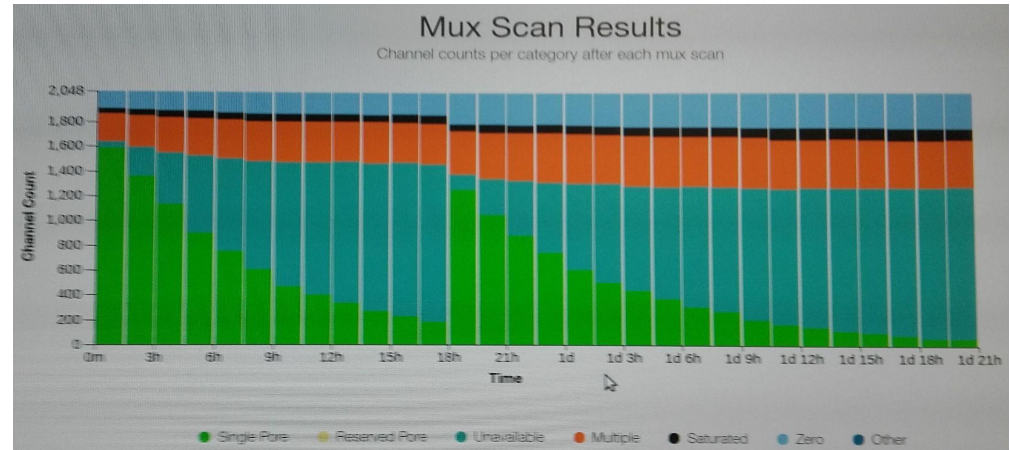
Flow cell performance over time

- Flow cell performance decreases over time
- Nanopores get blocked by DNA
- Nanopores break down due to air bubbles



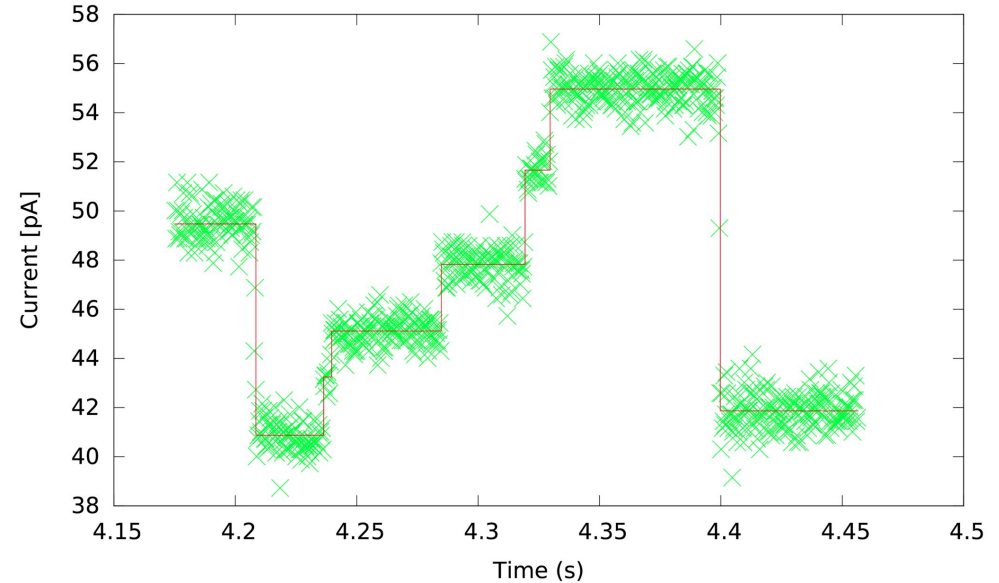
Stop & wash

- Sequencing is stopped once the number of active pores gets low
- Washing with DNase to free blocked nanopores
- Flow cells are regenerated and can be re-used
- Process can be repeated multiple times (3-5x)



Basecalling

- Electric signal is converted into sequence information (basecalling)
- Algorithmic improvement lead to higher read accuracy
- Raw sequencing data need to be stored

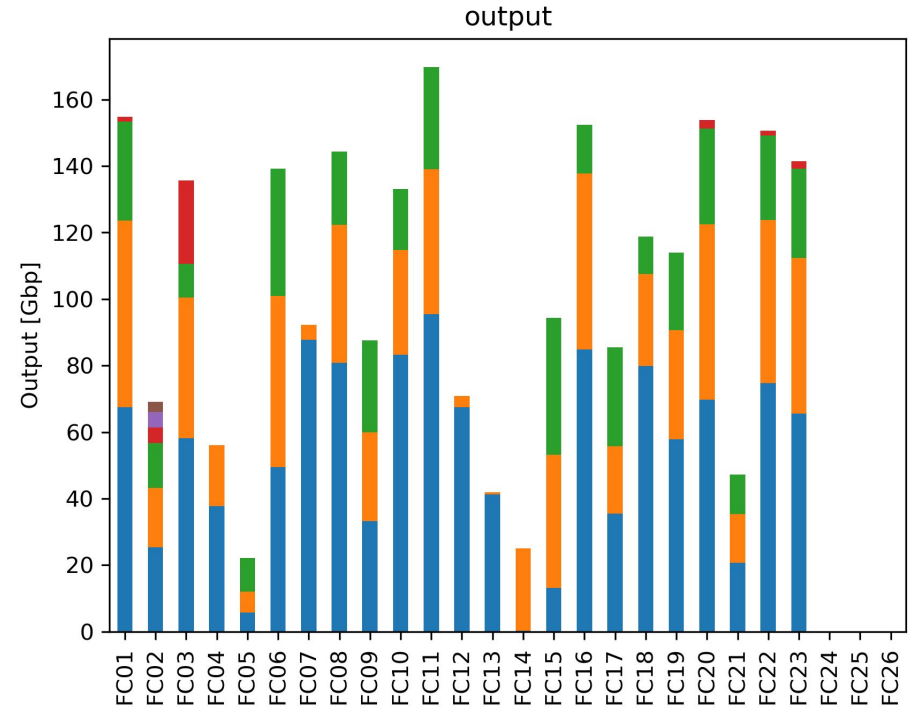


Basecalling with dorado

- Dorado allows conversion of electric signal into sequences
- Different models are available (HAC, SUP)
- Basecalling on CPUs is basically impossible
- Time estimates on GPUs: hours to days depending on dataset size

Calculating statistics

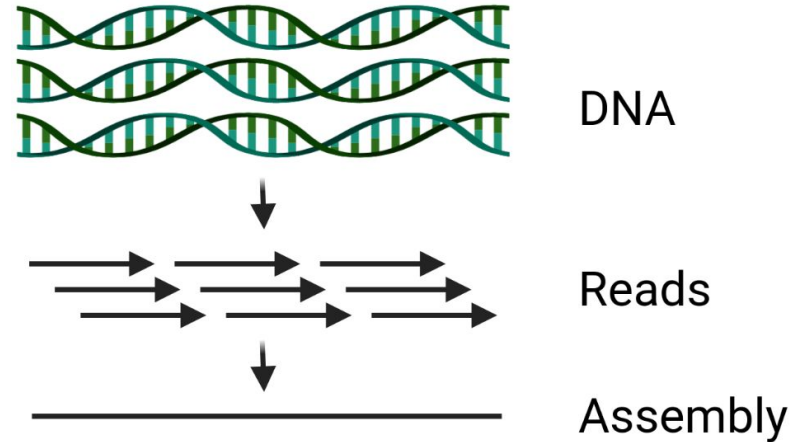
- Huge variation in output per flow cell
- Best runs: 150 Gbp
- Worst runs: 20 Gbp
- N50 for plants usually 25-65kb



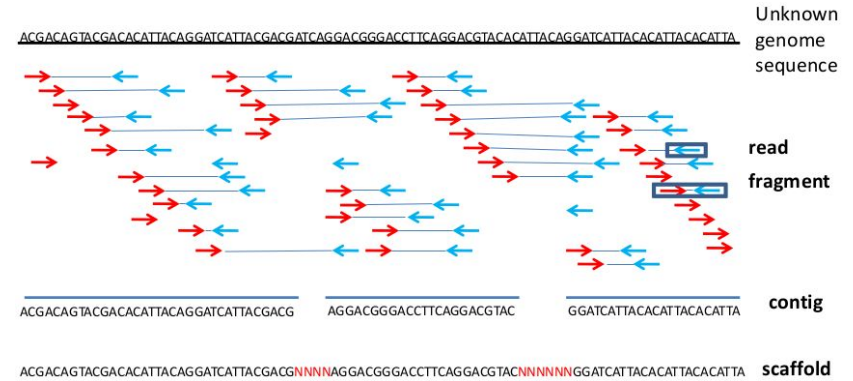
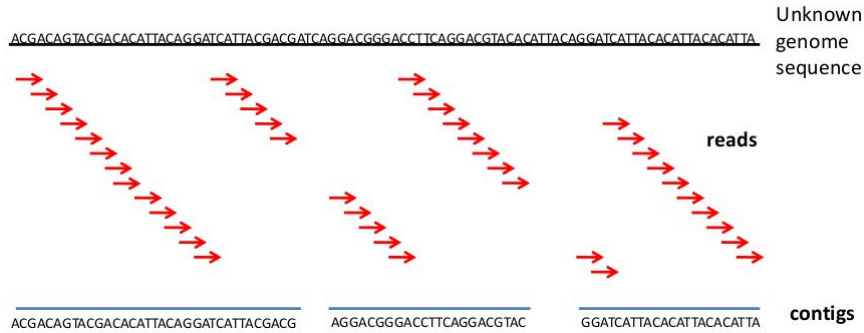
Genome Sequence Assembly

The assembly problem

- Reads are shorter than the chromosome
 - even long reads
- Multiple copies of the genome (DNA exist that can be subjected to sequencing
- Assembly = putting sequence pieces together (finding the common string of all substrings)
- Genome = DNA in a cell
- Genome sequence = representation of the DNA in a cell



Contigs, scaffolds, pseudochromosomes

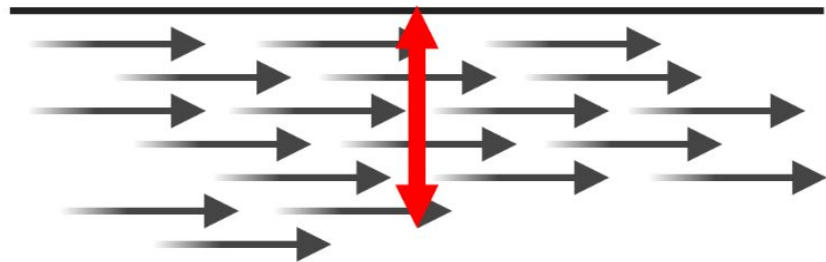


Pseudochromosomes = scaffolds representing an entire chromosome

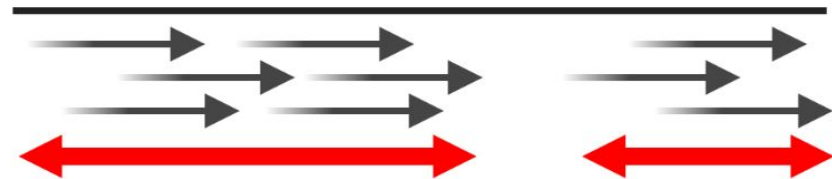
Gaps = regions between contigs that are represented by Ns

Sequencing coverage depth vs. coverage extent

Sequencing coverage depth

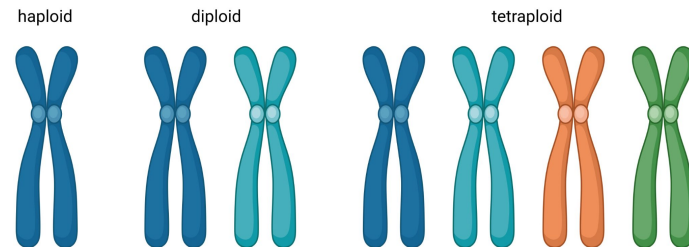
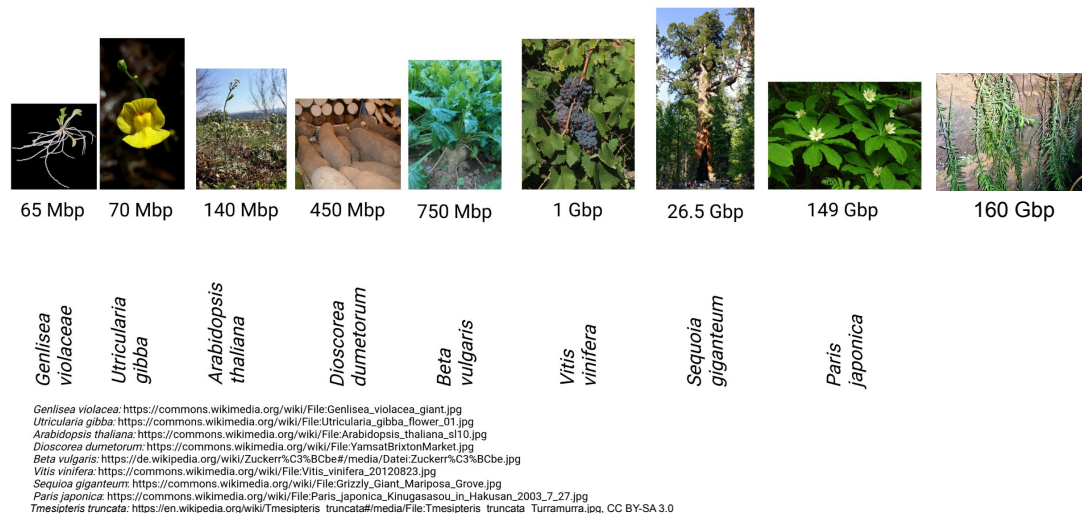


Sequencing coverage extent



Assembly challenges

- Genome size: variation from 65 Mbp to 160 Gbp
- Ploidy: haploid/diploid genomes are much easier to analyze than polyploid genomes



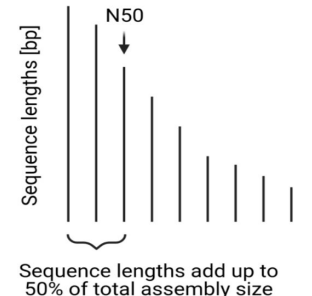
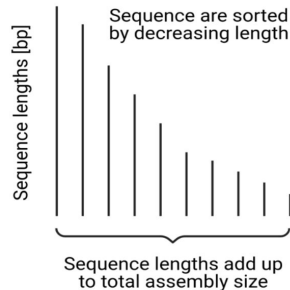
Assembly evaluation

- Continuity: Does the assembly represent a genome in a small number of contigs?
- Completeness: Are all parts of the genome represented?
- Correctness: Is the assembly a correct representation of the genome?

Evaluation: continuity

- Check assembly continuity: calculation based on sequences in FASTA file
- Number of contigs, assembly size, N50

assembler	Canu	FALCON	Miniasm	Flye
number of contigs	69	26	72	44
assembly size	123.5 Mbp	119.5 Mbp	120.2 Mbp	117 Mbp
maximal contig length	15.9 Mbp	15.9 Mbp	14.3 Mbp	14.9 Mbp
N50	13.4 Mbp	9.3 Mbp	8.6 Mbp	10.6 Mbp
N90	2.9 Mbp	2.8 Mbp	1.4 Mbp	2.5 Mbp



Evaluation - completeness

- Check assembly completeness: inspection for presence of conserved genes
- BUSCO = Benchmarking Universal Single-Copy Orthologue
- BUSCO genes are used to assess assembly completeness

BUSCO

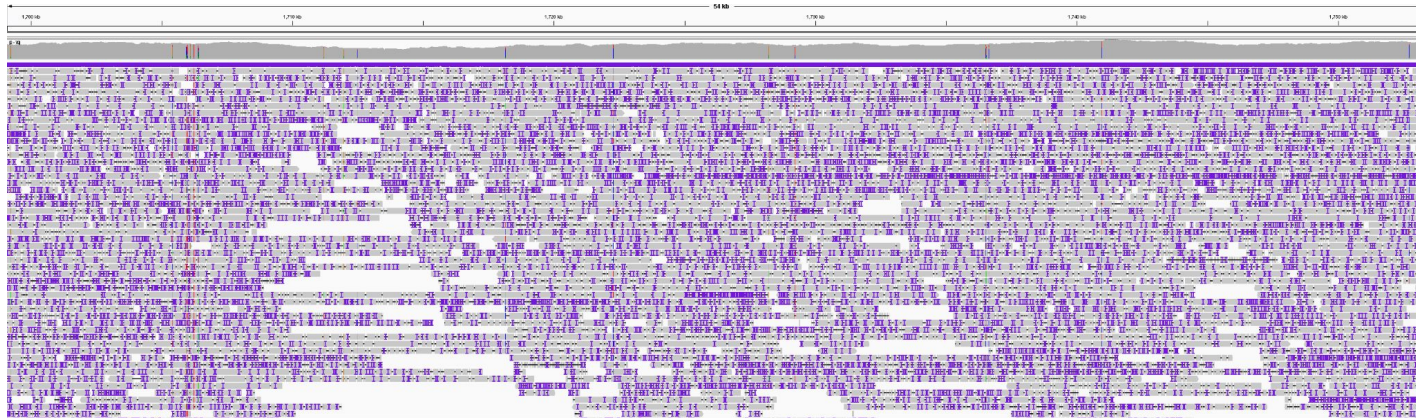
from QC to gene prediction and phylogenomics

BUSCO v5.3.2 is the current stable version!

[Gitlab](#), a [Conda package](#) and [Docker container](#) are also available.

Evaluation - assembly correctness

- Check assembly correctness: analyses of read mappings
- Integrative Genomics Viewer (IGV) can visualize read mappings
- Tools: REAPR, SQUAT

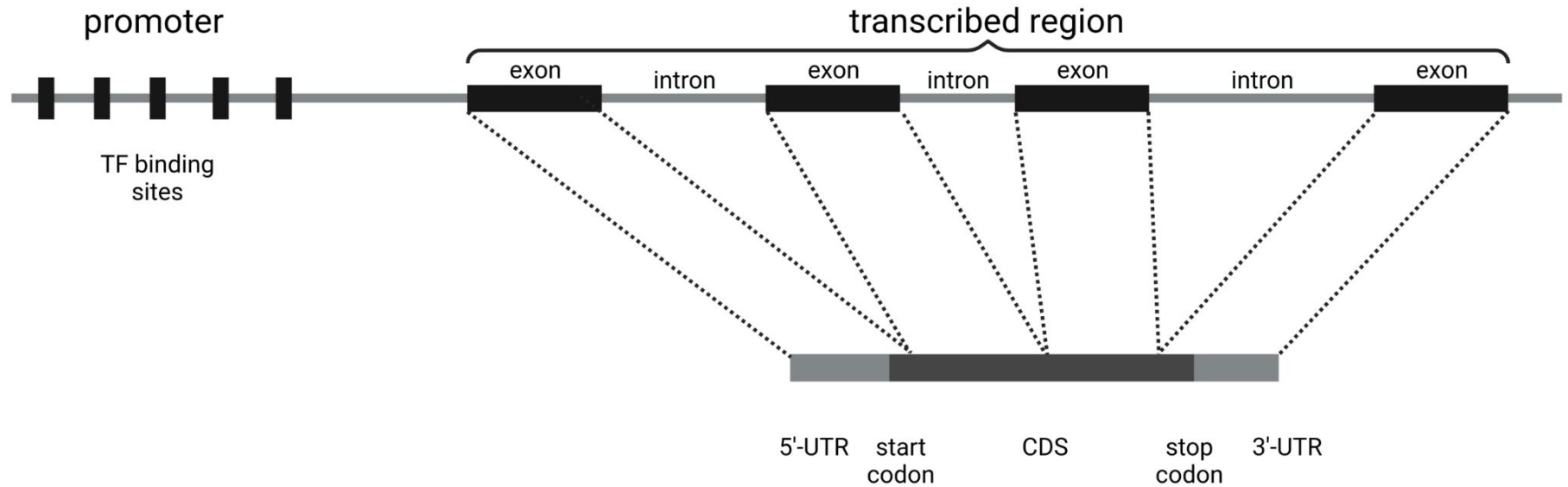


Structural annotation

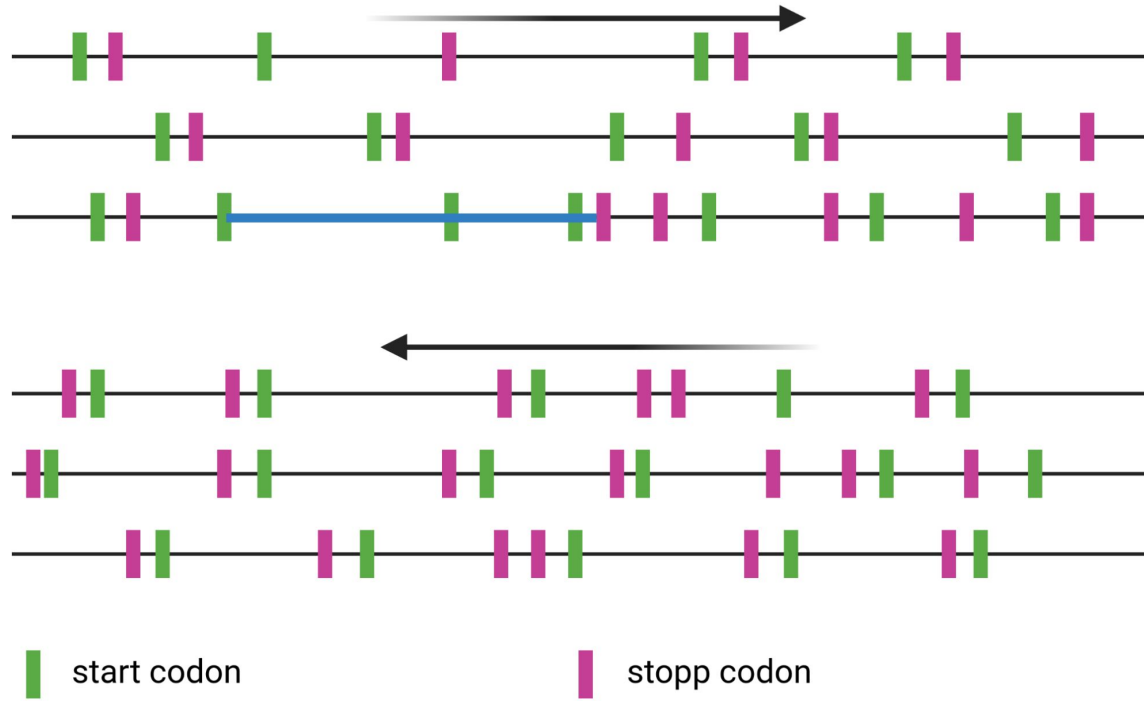
Plant gene structure

CDS = (Protein) Coding Sequence

ORF = Open Reading Frame

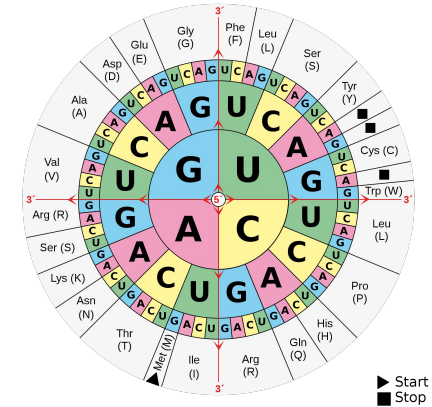


Finding ORFs

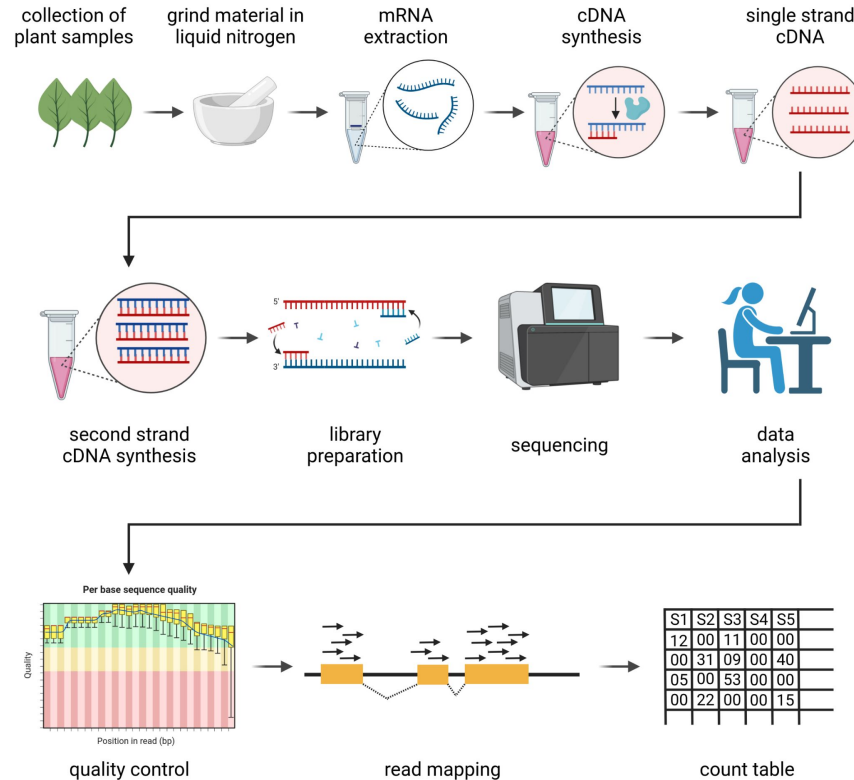


Codon usage

- Protein coding sequences have specific properties
- Codon usage is different between species i.e. different species prefer different codons for certain amino acids
- Rare codons slow down translation (useful between domains)
- Usage of dicodons (=hexamers)
- Codon usage can give additional CDS/ORF support

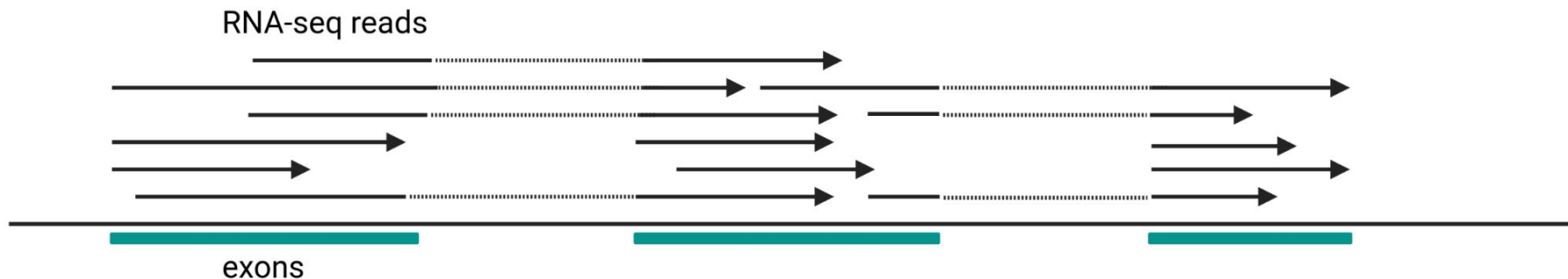


RNA-seq



RNA-seq hints

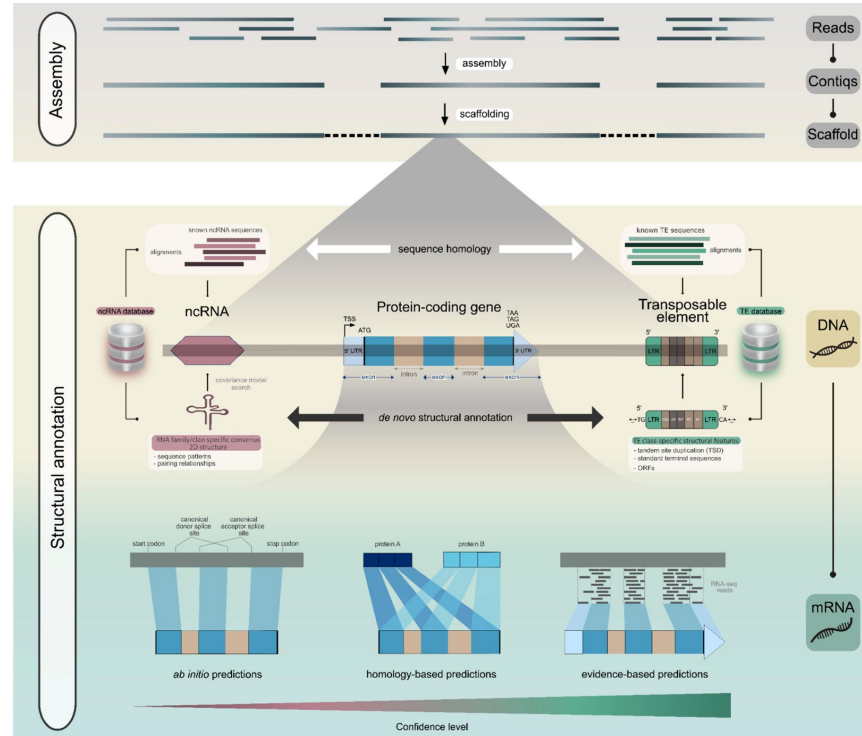
- Aligned RNA-seq reads indicate exon positions
- Splitting of reads indicates intron positions
- CDS can be identified as ORF within the covered regions



Annotation quality assessment

- BUSCO (Benchmarking Universal Single-Copy Orthologs)
 - Example: C:98%,S:85%,D13%,F:1.2%,M0.8%,n:1500
 - C = complete BUSCO genes
 - S = single copy BUSCO genes
 - D = duplicates BUSCO genes
 - F = fragmented BUSCO genes
 - M = missing BUSCO genes
 - n = number of BUSCO query sequences
- NLRome: assessment of high continuity genome sequences

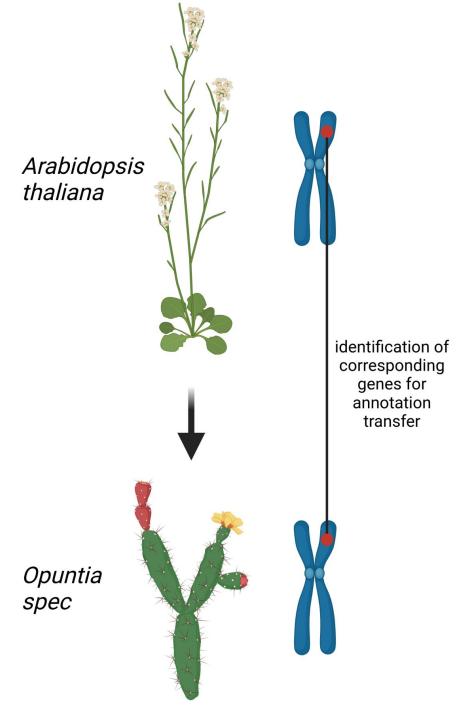
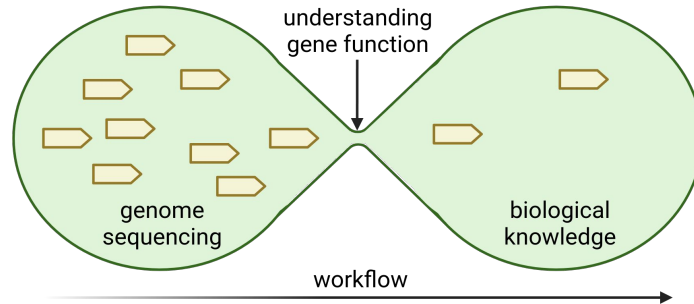
Structural annotation (overview)



Functional annotation

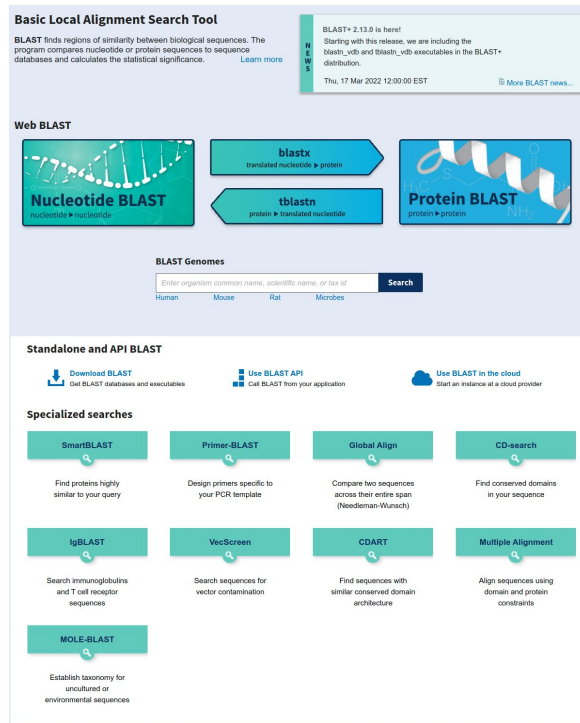
Functional annotation

- Assumption: similar sequences (genes) have similar functions
- Sequence similarity indicates origin from a shared common ancestor
- Ancestor is likely to have inherited the same function to offspring
- Transfer of functional annotation based on sequence similarity
- *Arabidopsis thaliana* gene functions are well studied and usually serve as reference



BLAST: Basic Local Alignment Search Tool

- Probably the most famous website of the NCBI
- Comparison of sequences against a large database
- Similar sequences are likely to have similar functions (ideally orthologs)
- Numerous variants of the initial BLASTn were developed



The screenshot shows the NCBI BLAST homepage. At the top, it says "Basic Local Alignment Search Tool" and provides a brief description of BLAST's function. A news banner for BLAST+ 2.13.0 is visible. The main section is titled "Web BLAST" and features two primary search options: "Nucleotide BLAST" (nucleotide vs nucleotide) and "Protein BLAST" (protein vs protein). Below these, there are buttons for "blastx" (translated nucleotide vs protein) and "tblastn" (protein vs translated nucleotide). A "BLAST Genomes" section allows users to select an organism (Human, Mouse, Rat, Mice) and enter a search query. The "Standalone and API BLAST" section offers options to download BLAST, use the BLAST API, or use BLAST in the cloud. The "Specialized searches" section includes links to SmartBLAST, Primer-BLAST, Global Align, CD-search, IgBLAST, VecScreen, CDART, Multiple Alignment, and MOLE-BLAST, each with a brief description of its purpose.

- Screen of protein sequences against collection of protein signatures
- Allows the assignment of functional annotation terms
- Available as web service, but also as stand alone tool

Results for job iprscan5-I20220320-102557-0980-87120812-p2m

Tool Output		Submission Details																					
Download in XML format				Download in TSV format				Download in GFF3 format				Download in SVG format				Download HTML tarball file				Download in JSON format			
sp	P13114	CHSY_ARATH	ba46b0f06bfbf1c91d161f0f91310f43	395	Gene3D	G3DSA:3.40.47.10	-	1	241	4.8E-101	T	20-03-2022											
sp	P13114	CHSY_ARATH	ba46b0f06bfbf1c91d161f0f91310f43	395	SUPERFAMILY	SSF53901	Thiolase-like	241	393	2.98E-51	T	20-03-2022											
sp	P13114	CHSY_ARATH	ba46b0f06bfbf1c91d161f0f91310f43	395	SUPERFAMILY	SSF53901	Thiolase-like	10	237	2.38E-78	T	20-03-2022											
sp	P13114	CHSY_ARATH	ba46b0f06bfbf1c91d161f0f91310f43	395	Pfam	PF02797	Chalcone and stilbene synthases, C-terminal domain				244	394	1.5E-71										
sp	P13114	CHSY_ARATH	ba46b0f06bfbf1c91d161f0f91310f43	395	PANTHER	PTHR11877:SF81	BNAA02G30320D PROTEIN	6	394	0.0	T	20-03-2022											
sp	P13114	CHSY_ARATH	ba46b0f06bfbf1c91d161f0f91310f43	395	Gene3D	G3DSA:3.40.47.10	-	242	395	9.7E-62	T	20-03-2022											
sp	P13114	CHSY_ARATH	ba46b0f06bfbf1c91d161f0f91310f43	395	PANTHER	PTHR11877	HYDROXYMETHYLGLUTARYL-COA SYNTHASE				6	394	0.0	T	20-03-2022								
sp	P13114	CHSY_ARATH	ba46b0f06bfbf1c91d161f0f91310f43	395	CDD	cd00831	CHS_like	21	390	0.0	T	20-03-2022											
sp	P13114	CHSY_ARATH	ba46b0f06bfbf1c91d161f0f91310f43	395	PIRSF	PIRSF000451	PKS III	7	394	0.0	T	20-03-2022											
sp	P13114	CHSY_ARATH	ba46b0f06bfbf1c91d161f0f91310f43	395	Pfam	PF00195	Chalcone and stilbene synthases, N-terminal domain				10	233	2.9E-119										
sp	P13114	CHSY_ARATH	ba46b0f06bfbf1c91d161f0f91310f43	395	ProSitePatterns	PS00441	Chalcone and stilbene synthases active site.				161	177	-										

InterProScan 5 Sequence Search

This form allows you to scan your sequence for matches against the InterPro collection of protein signature databases.

This form is for debugging purposes only and is **not supported**. To submit jobs to InterProScan 5, please visit the [InterPro Sequence Search](#) or the [InterProScan 5 Web services](#).

Please Note

This web form is for internal debugging purposes and will be retired in April 2016. If you have any questions/concerns please contact us via the feedback link above.

STEP 1 - Enter your input sequence

Enter or paste a **PROTEIN** sequence in any supported format:

uniprot:KPXM_HUMAN

Or, upload a file: No file chosen

[Use a sample sequence](#) | [Clear sequence](#) | [See more example inputs](#)

STEP 2 - Select the applications to run

☒ TIGRfam ☒ RFLD ☒ Probus ☒ SignalP ☒ SignalP_Euk
☒ SignalP_Gram_Positive ☒ SignalP_Gram_Negative ☒ SUPERFAMILY ☒ PANTHER ☒ Gene3D
☒ Pfam ☒ ProSiteProfiles ☒ ProSitePatterns ☒ CDD ☒ SMART
☒ CDD ☒ PRINTS ☒ Pfam ☒ Pfam-CLUSTal ☒ PIRSF

STEP 3 - Submit your job

☐ Be notified by email (Tick this box if you want to be notified by email when the results are available)

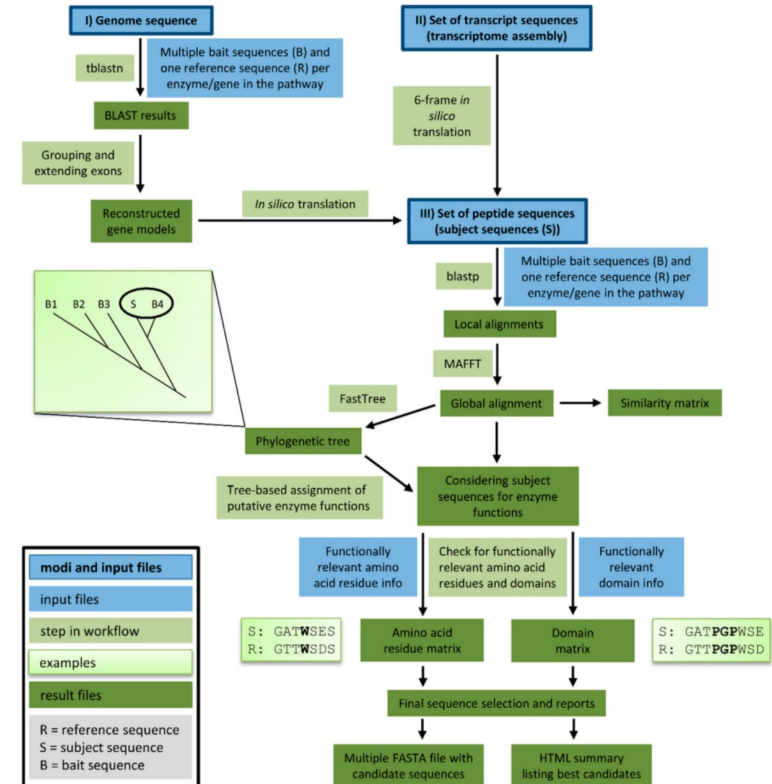
Gene Ontology (GO)

- Defined statements about the function of a gene (controlled vocabulary)
- Hierarchical structure
 - Example: 'metabolic process' > 'biosynthetic process' > ... > 'chalcone synthase'
- Supported by the Alliance of Genome Resources
- Connected with various other databases e.g. TAIR, FlyBase, Reactome, UniProt
- Machine readable to allow automatic processing
- Tools: Blast2GO and AmiGO
 - analyze the function of a sequence (web service and standalone)

```

1 GO:0008150 biological_process
1 GO:0008152 metabolic process
1 GO:0009058 biosynthetic process
1 GO:0071704 organic substance metabolic process
1 GO:0009812 flavonoid metabolic process
1 GO:1901576 organic substance biosynthetic process
  GO:0009813 flavonoid biosynthetic process
    1 GO:0009718 anthocyanin-containing compound biosynthetic process
    1 GO:0051551 aurone biosynthetic process
    1 GO:0033485 cyanidin 3-O-glucoside biosynthetic process
    1 GO:0033486 delphinidin 3-O-glucoside biosynthetic process
    1 GO:0051553 flavone biosynthetic process
    1 GO:0009716 flavonoid phytoalexin biosynthetic process
    1 GO:0051557 leucoanthocyanidin biosynthetic process
    R GO:0009964 negative regulation of flavonoid biosynthetic process
    1 GO:0033487 pelargonidin 3-O-glucoside biosynthetic process
    G GO:0009963 positive regulation of flavonoid biosynthetic process
    R GO:0009962 regulation of flavonoid biosynthetic process
  
```

- Identification of all players in a biosynthesis pathway
- Dedicated to the identification of enzymes
- Functionally relevant amino acids need to be known
- Web server: <http://www.pbb-tools.de>



Databases for submission of sequencing data

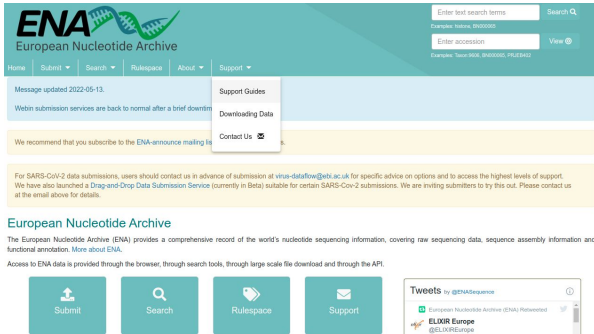
- Sequence Read Archive



SRA - Now available on the cloud

Sequence Read Archive (SRA) data, available through multiple cloud providers and NCBI servers, is the largest publicly available repository of high throughput sequencing data. The archive accepts data from all branches of life as well as metagenomic and environmental surveys. SRA stores raw sequencing data and alignment information to enhance reproducibility and facilitate new discoveries through data analysis.

- European Nucleotide Archive



ENA
European Nucleotide Archive

Enter text search terms Search

Enter accession View

Home Submit Search Rulespace About Support

Message updated 2022-05-13.
Webin submission services are back to normal after a brief downtime.

We recommend that you subscribe to the ENA-announce mailing list.

For SARS-CoV-2 data submissions, users should contact us in advance of submission at china-dataflow@ebi.ac.uk for specific advice on options and to access the highest levels of support. We have also launched a [Drag-and-Drop Data Submission Service](#) (currently in Beta) suitable for certain SARS-CoV-2 submissions. We are inviting submitters to try this out. Please contact us at the email above for details.

European Nucleotide Archive

The European Nucleotide Archive (ENA) provides a comprehensive record of the world's nucleotide sequencing information, covering raw sequencing data, sequence assembly information and functional annotation. [More about ENA.](#)

Access to ENA data is provided through the browser, through search tools, through large scale file download and through the API.

Submit Search Rulespace Support

Tweets by [@genomics_eu](#)

European Nucleotide Archive (ENA) Powered by **ELIXIR Europe**

Nanopore sequencing applications

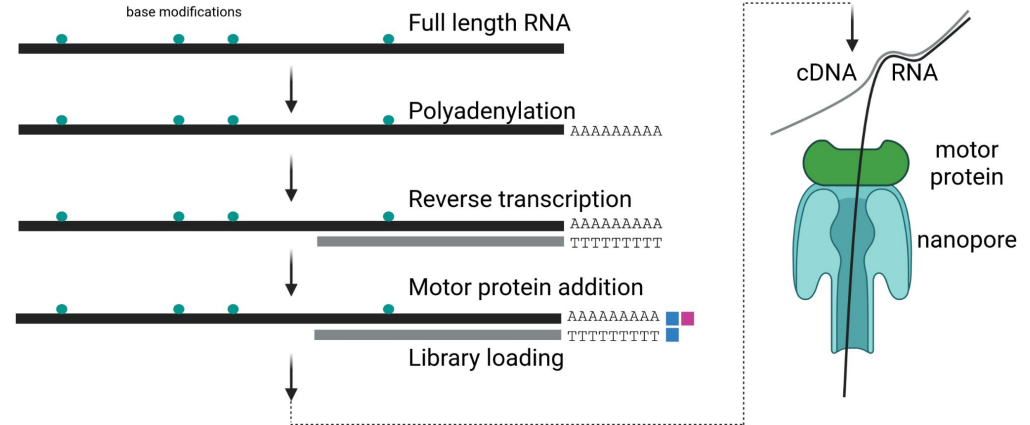
Adaptive sampling

- Check DNA strand based on first sequenced bases against reference sequence:
 - strand is of interest: continue
 - strand is not of interest: reject
 - Example: only sequence bacterial/fungal DNA included in a plant sample

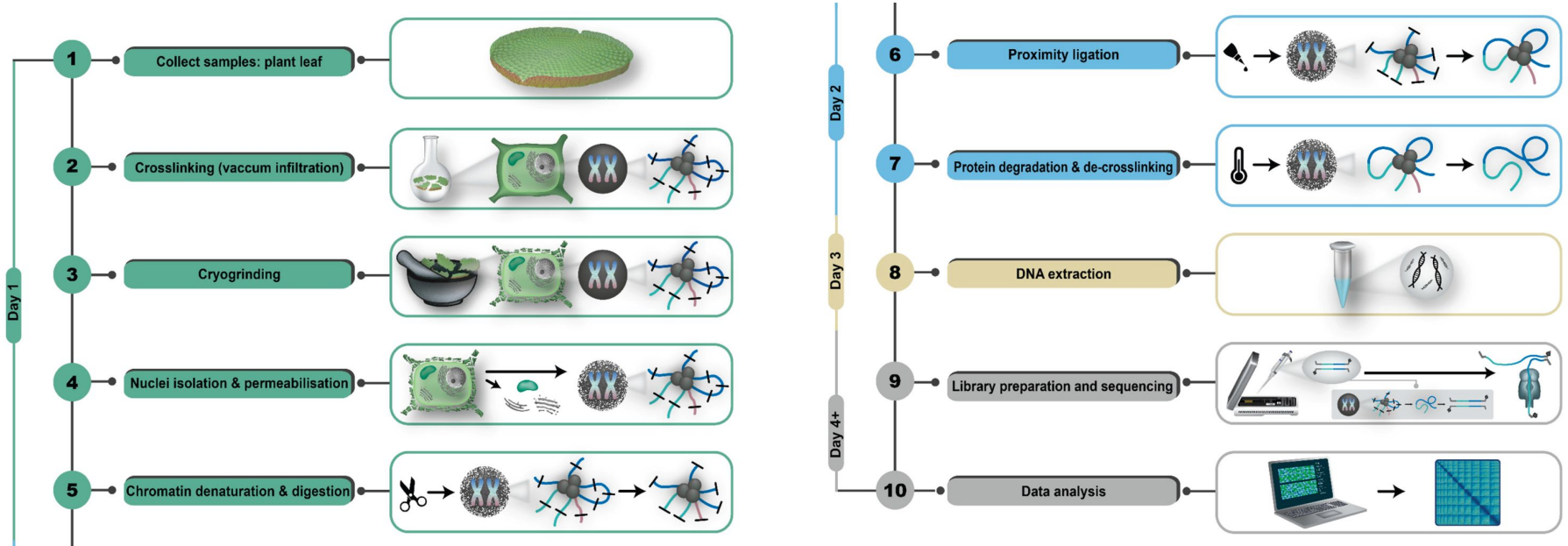


Direct RNA sequencing

- PolyA tail (3') required for primer ligation
- Generation of complementary DNA strand through reverse transcriptase recommended
- Sequencing of RNA strand allows analysis of base modifications

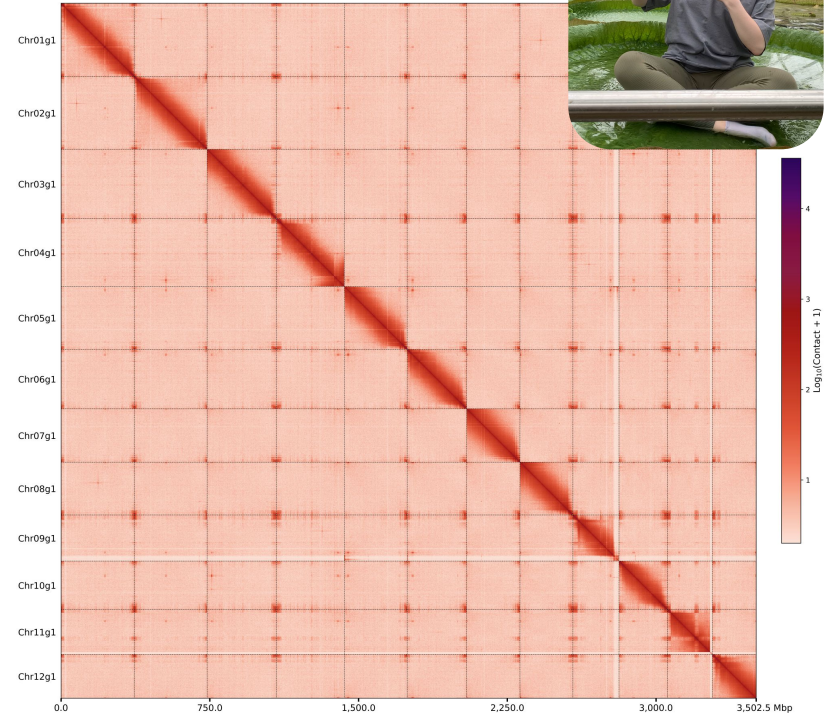
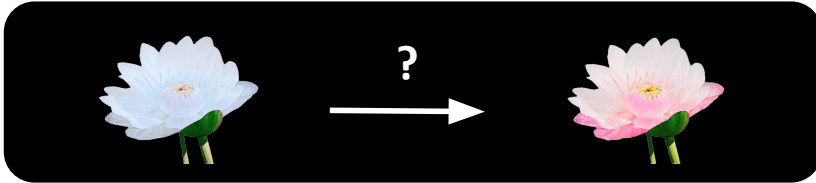


Pore-C (chromatin structure)



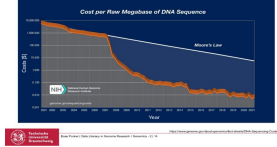
Victoria cruziana

- Water lily with giant floating leaves
- Flowers at night and floral color transitions from white to pink
- Pore-C resulted in an almost perfect genome sequence

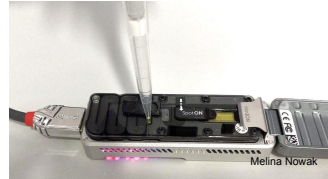
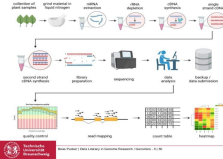


Teaching sequencing in practical courses

Development of sequencing capacity



RNA-Seq



Data Literacy and
Plant Genomics

Summary

- Sequencing methods: Sanger, Illumina, PacBio, ONT
- Planning the sequencing experiment
- DNA extraction, quality control, library preparation
- Nanopore sequencing & real time monitoring
- Genome sequence assembly
- Structural and functional annotation
- Nanopore sequencing applications

Erasmus opportunity

- Agreement between University of Thessaly and University of Bonn
- You could visit Bonn for a semester starting spring 2027
- Opportunity to take elective modules



Time for questions!

- de Oliveira, J. A. V. S.; Choudhary, N.; Meckoni, S. N.; Nowak, M. S.; Hagedorn, M.; Pucker, B. (2026). Cookbook for Plant Genome Sequences. doi:10.1186/s12864-026-12623-z.
- Wolff, K.; Friedhoff, R.; Schwarzer, F.; Pucker, B. (2023). Data Literacy in Genome Research. Journal of Integrative Bioinformatics, 2023, pp. 20230033. doi: 10.1515/jib-2023-0033.
- Pucker B, Irisarri I, de Vries J and Xu B (2022). Plant genome sequence assembly in the era of long reads: Progress, challenges and future directions. Quantitative Plant Biology, 3, E5. doi: 10.1017/qpb.2021.18.